

Introduction to NC State's HPC (Hazel)

*A researcher considering
shared resources needs to
consider these 2 things*

- **Quick GPU access to get started**
- **10x — scaling up for higher throughput**
- **Local `claude.code` demo**
- **Your turn (workshop segment)**

Quick Access

10x

Local Claude

Your Setup

Quick Access: SLURM & LSF

Two ways to find and request GPU queues

SLURM quick access

```
# to see available nodes  
si --nodes --gpus  
  
--partition=gpu_partners --qos=short_gpu
```

LSF quick access

```
bqueues -l short_gpu  
bhosts -X gpu_l40s  
  
-q short_gpu  
-R "select[l40s|a100|a30|h100|h200]"
```

Ask HPC Assistant

Quick Access

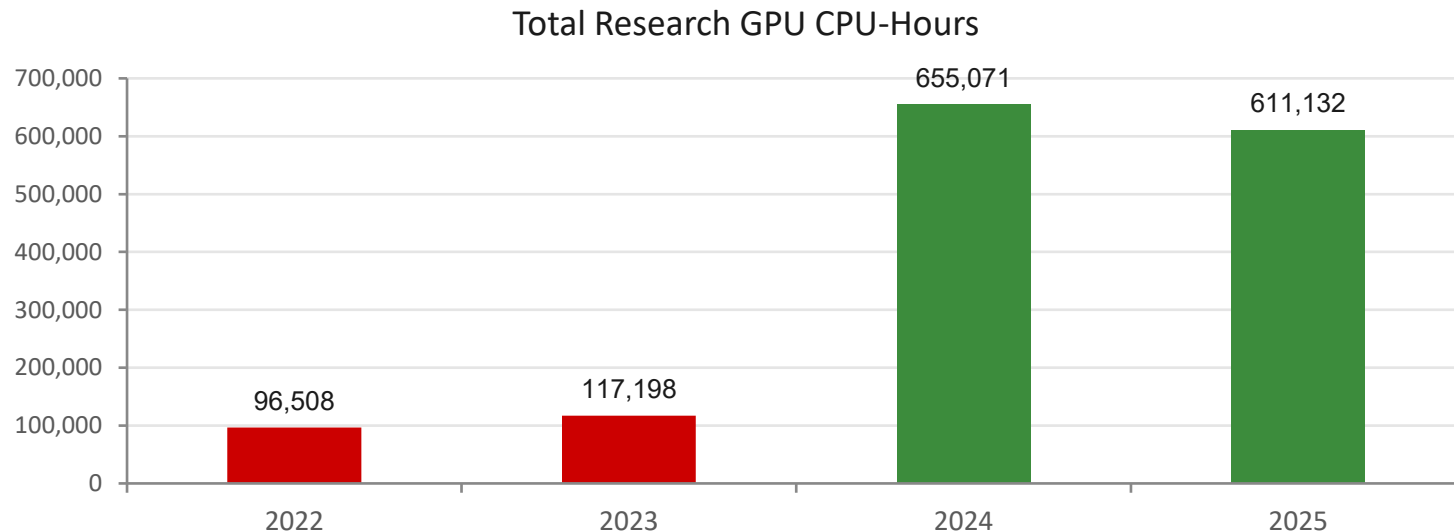
10x

Local Claude

Your Setup

GPU Usage: Research Projects

Total research GPU CPU-hours by year



Quick Access

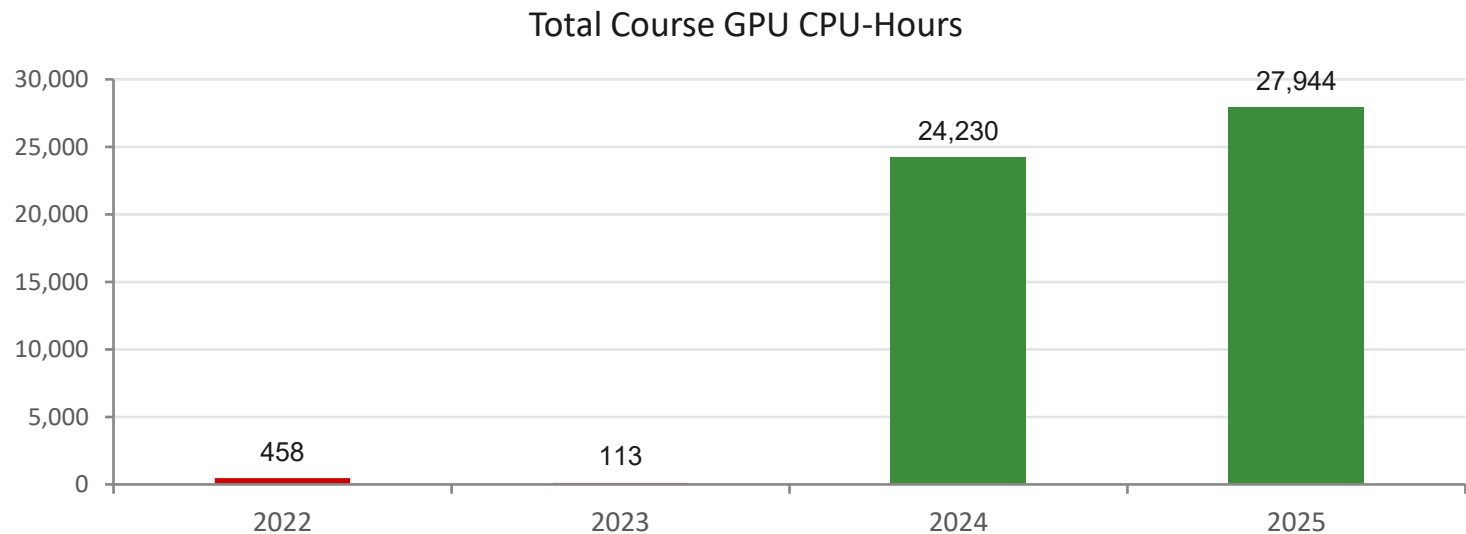
10x

Local Claude

Your Setup

GPU Usage: Coursework

Total course GPU CPU-hours by year



Quick Access

10x

Local Claude

Your Setup

Why such a big jump in teaching?

New capacity and integrations aimed squarely at coursework

5TB of /share

Scratch space that is NOT wiped

Open OnDemand

Class/teaching or research applications integrated directly into OOD

Browser login without password

15GB in /home/\$USER

Per-user home directory storage

HPC-VCL server reservation

Run a service — like SQL or an LLM/RAG client — for class or research

Quick Access

10x

Local Claude

Your Setup

Current capacity and growth projections

GPU inventory available today

Current

H200	H100	L40S	A100 / A30 / A10
12	12	60	20

Growth Projections

Additional GPU capacity coming online

Timeline	Additional GPUs	Type
2 months	8	H200
2 months	16	L40S
6 months	24	RTX 6000 Pro Blackwell
9 months	72	A100

Partner requests from researchers, departments, or centers — we can accept millions of \$ worth of equipment.

[Quick Access](#)[10x](#)[Local Claude](#)[Your Setup](#)

Integration with Core Facilities

Research Storage

HPC-VCL

Quick Access

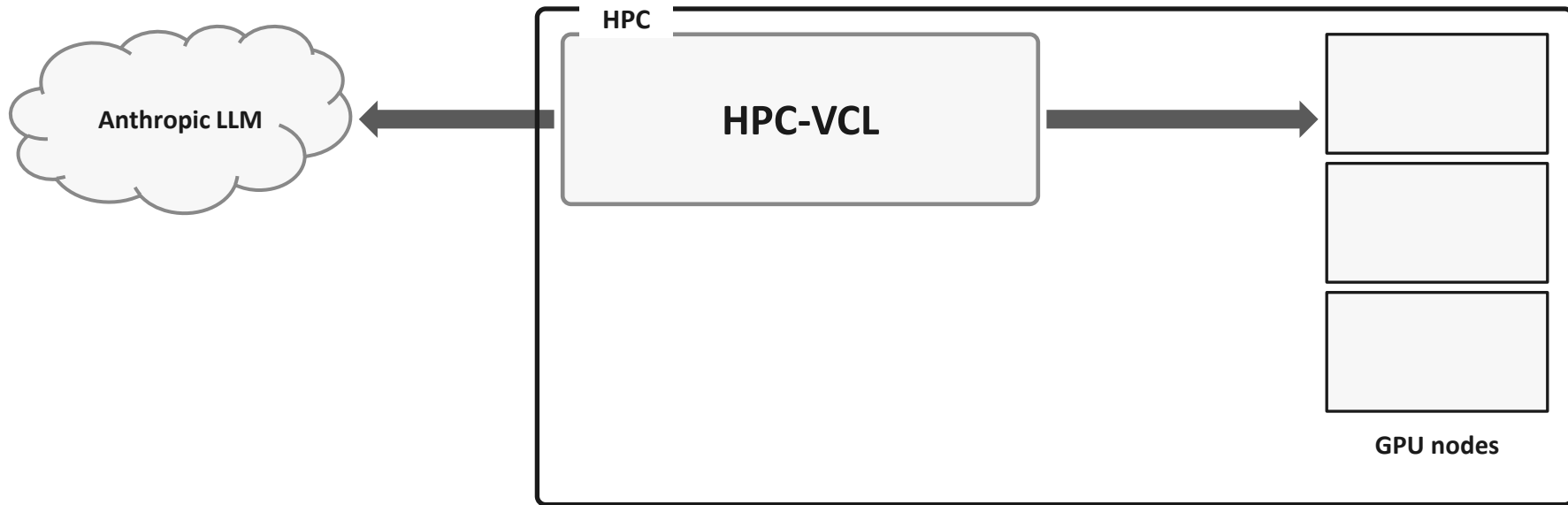
10x

Local Claude

Your Setup

Local claude.code Demo

Same client, two different backends



Quick Access

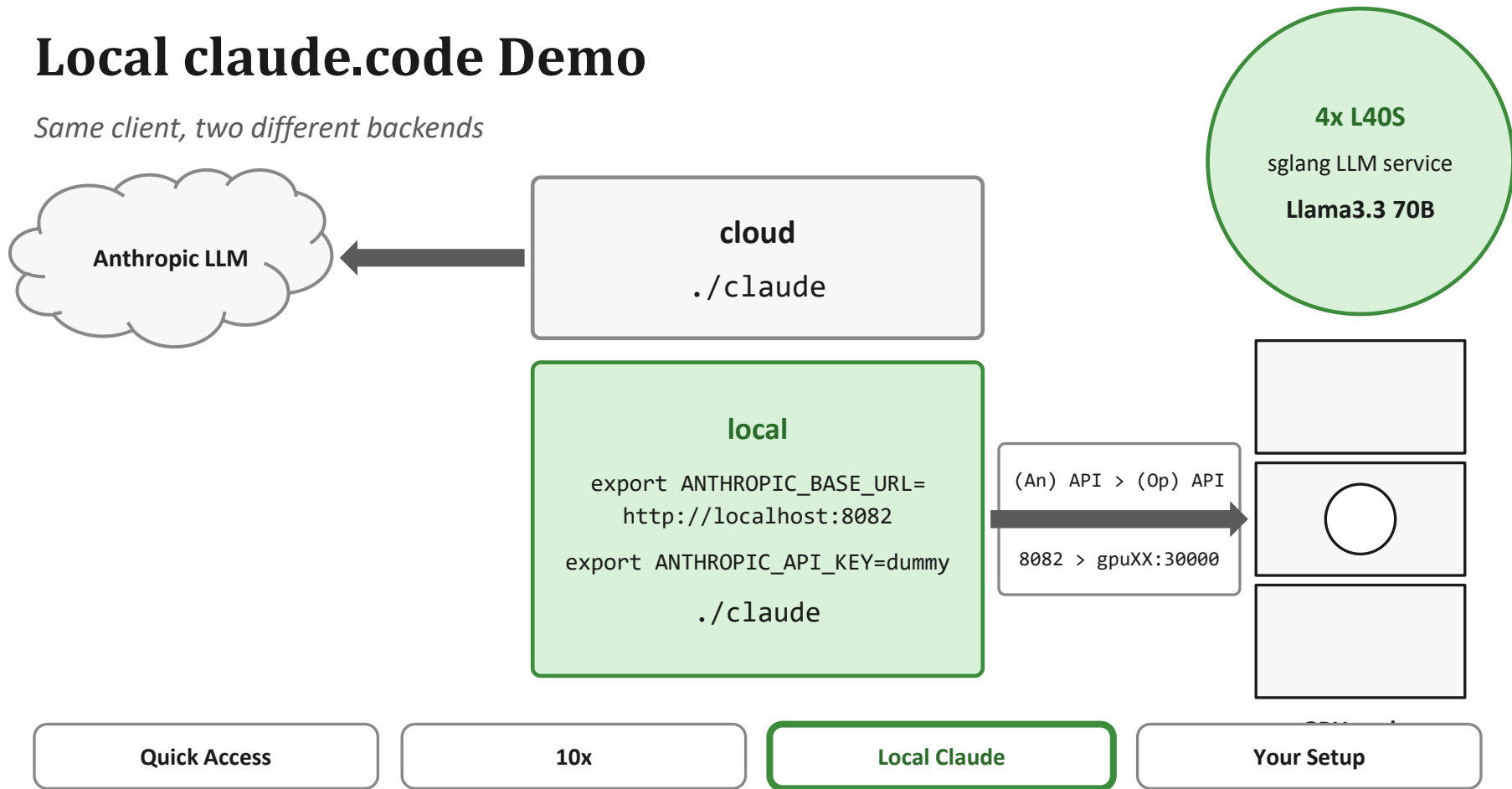
10x

Local Claude

Your Setup

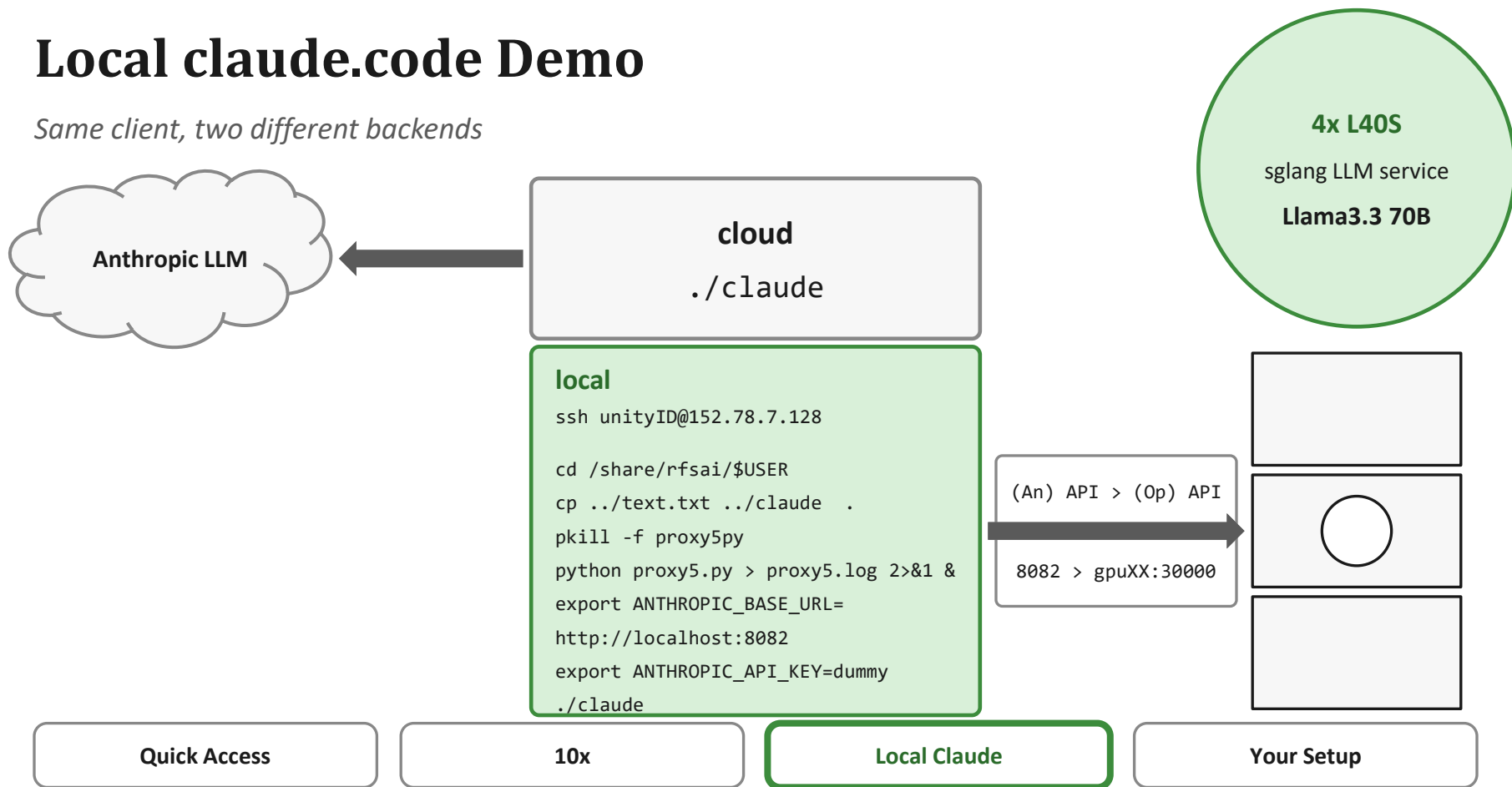
Local claude.code Demo

Same client, two different backends



Local claude.code Demo

Same client, two different backends



Your setup

local

```
ssh unityID@152.78.7.128
```

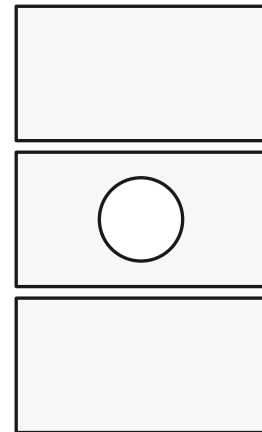
```
#-----LiteLLM-----
```

```
export ANTHROPIC_BASE_URL=http://localhost:8082
```

```
export ANTHROPIC_API_KEY=dummy
```

```
./claude
```

```
Fix the spelling in the file ./text.txt
```



GPU nodes

Quick Access

10x

Local Claude

Your Setup

GPU Requirements for Open-Source Models

How many GPUs it takes to run models that compete with Anthropic's LLMs

Model	Parameters	H200 (141GB)	H100 (80GB)	L40S (48GB)	Tool Use Quality
DeepSeek-V3	671B MoE	8x	16x	<i>not practical</i>	★★★★★ closest to Sonnet
Kimi K2	~1T MoE	8x+	<i>not practical</i>	<i>not practical</i>	★★★★★
Qwen3-235B-A22B	235B MoE	2x	4x	<i>not practical</i>	★★★★
Mistral Small 4	119B MoE	1x	1x	<i>not practical</i>	★★★★
Qwen3-32B	32B dense	1x	1x	1x	★★★
Llama3.3 70B (current)	70B dense	1x	1x	2x	★★★

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HPC History

Before NC State: the early years of the NC Supercomputing Center

Pre-2003

- Housed at the NC Supercomputing Center
- System developed in partnership with IBM
- Ranked among the top 20 supercomputers worldwide
- November 2000: ranked #16 on the Top500 list

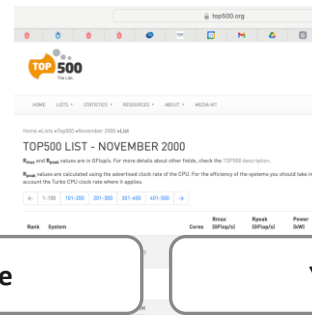
Quick Access

10x

Local Claude

Your Setup

15	SP Power3 375 MHz, IBM Charles Schwab United States	768	795.00	1,152.00
16	SP Power3 375 MHz, IBM North Carolina Supercomputing Center (NCSC) United States	720	741.00	1,080.00
17	SP Power3 375 MHz, IBM DOE/SC/Oak Ridge National Laboratory United States	704	723.00	1,056.00



It Works

Claude Code fixing the typo in text.txt, end to end

```
aapeters — * Fix spelling in text file — ssh -Y aapeters@login.hpc.ncsu.edu — 95x32

What's new
ated": 1783548350, "mod n2.5", "choices": [{"inAdded an auto mode rule that blockstt..t",
, "content": "{\\type n\\", \\name\\: \\\"ReadFixed `--json-schema` silently produc"/r
sssglang response: {"id 08f19644ec0bb2f71a46051Fixed a message sent while Claude was..re
ated": Opus 4.8 (1M context) · API Usage Billing: [{"in/release-notes for more " i t"
, "c/rsstu/users/a/aapeters/gpu-expansion/fineTuneile\\": 1 1 al

⚠ Both claude.ai and ANTHROPIC_API_KEY set · auth may not work as expected
· to use claude.ai: Unset the ANTHROPIC_API_KEY environment variable, or claude /logout then
, ssay "No" to the API key approval before login.
ate · to use ANTHROPIC_API_KEY: claude /logout to sign out of claude.ai.e": {"role": "assistant"
, "
Fix the spelling in /rsstu/users/a/aapeters/gpu-expansion/fineTune/text.txt

• Update(text.txt)

Edit file
text.txt

1 GPUs are greate
1 GPUs are great

Do you want to make this edit to text.txt? ade0c316ab2405e7", "object": "chat.completion", "cre
a) e1. Yes 83548359, "model": "qwen2.5", "choices": [{"index":
2. Yes, allow all edits during this session (shift+tab)
t3. Nors/a/aapeters/gpu-expansio

Esc to cancel · Tab to amend
```

Quick Access

10x

Local Claude

Your Setup